

Report on AD4630-24 EVAL Boards characterization

Ricardo Iuzzolino¹ and Luis Palafox²

¹Instituto Nacional de Tecnología Industrial (INTI) - Argentina

²Physikalisch-Technische Bundesanstalt (PTB) - Germany

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Abstract

This report enumerates different tests made at PTB for two AD4630-24 evaluation boards, one belongs to Left Right and the second to INTI. The main characterization was in terms of the Total Harmonic Distortion (THD) at different input amplitudes, then the results were compared with the specifications on the ADC datasheet. Both boards were synchronized using the same 100 MHz clock signal, allowing averaging on time-domain -channels parallel operation and averaging- and in the frequency domain -Fourier transform averaging-. Also, clock spurs frequencies were detected.

1 Introduction

Two AD4630-24 evaluation boards [1] were evaluated in terms of total harmonic distortion and parallel operation and averaging. The use of high resolution ADC is increasing, the main goal of the True8Digit project is to show the possibility to go up to 8-1/2 digit exiting signal averaging and ADC parallel operation.

2 Total Harmonic Distortion Test and Analysis

2.1 Measurement Setup

The measuring setup is shown in figure 1, the ultra low distortion source, the APx555 was used to excite the ADC input. Two AD4630-24 evaluation boards were connected in parallel using a custom connection box, as can be seen in the center of figure 1). In order to synchronize both ADCs, a master/slave configuration was set, the main clock of RA EVAL boards (right-top of fig.1) of frequency equal to 100 MHz was connected to RI (right-bottom of fig.1) board as is depicted in the figure by the blue 3.5 mm cable.

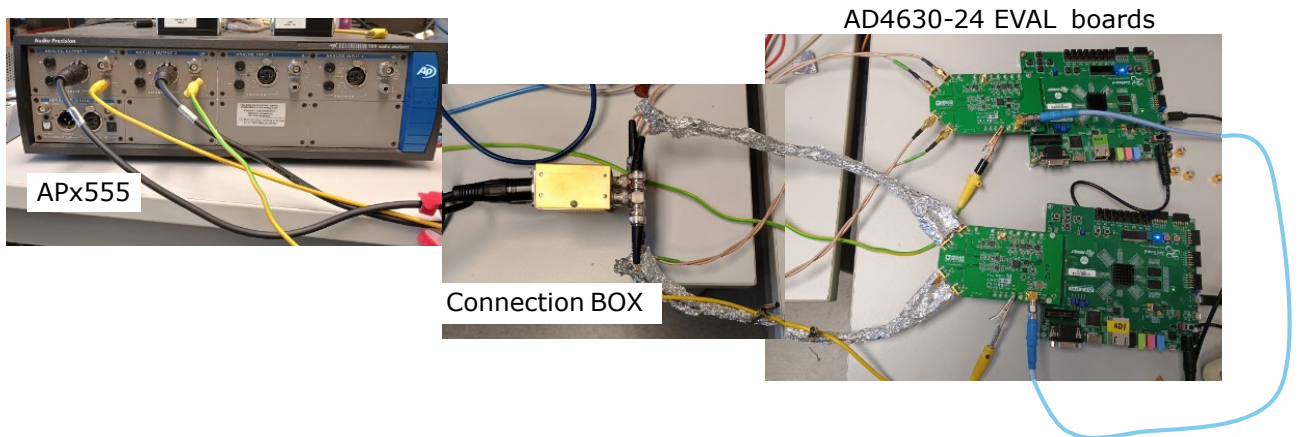


Figure 1: Measurement setup. The APx55 deliver a low distorted signal, a parallel connection was possible using a connection box (middle) and the two evaluation boards running with the same clock, RA board at the right-top as master and the RI board right-bottom as slave.

2.2 Two board analysis

The first study was to compute the Total Harmonic Distortion of the two boards versus input signal amplitudes and frequency, taken into account the first fifth harmonics as the ratio between the squared root of the square sum of the harmonic to the RMS voltage of the fundamental frequency as indicated by equation 1.

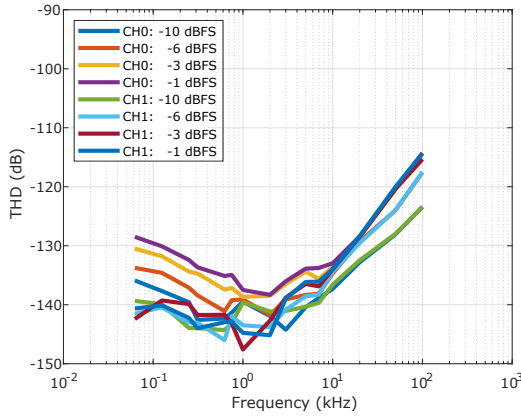
$$THD = \frac{\sqrt{\sum_{j=2}^5 V_{RMSj}^2}}{V_{RMS0}} \quad (1)$$

The RMS voltages were obtained in the frequency domain using a windowed FFT, window applied to the ADC data was a 7-term Blackman-Harris [2].

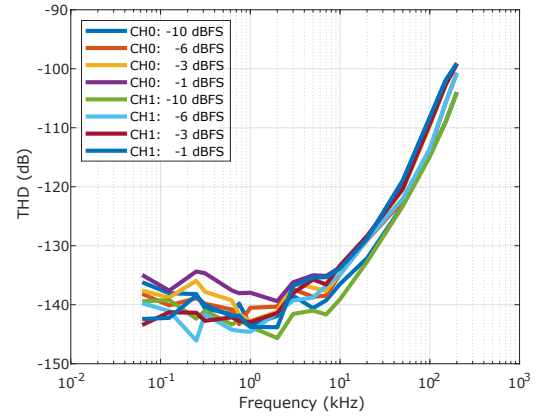
Amplitudes range were: -10 dBFS, -6 dBFS, -3 dBFS and -1 dBFS, at frequencies from 62.5 Hz up to 200 kHz. The results are shown in Table 1 for RI board and plotted in figure 2a for RA board and in figure 2b for RI board.

THD versus input amplitudes and frequencies for the two boards under test are depicted in figure 2 and table 1

Figure 2: Board RA and RI - THD vs Input signal amplitude and frequency



(a) Board RA - THD



(b) Board RI - THD

Table 1: Board RI - Results

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3274	-136.1918	90.7905	16.50
124.9	-10.3274	-138.0479	90.7806	16.50
250.1	-10.3274	-138.1932	90.7350	16.50
312.6	-10.3274	-140.6363	90.7447	16.50
625.1	-10.3275	-141.5769	90.7245	16.49
750.1	-10.3275	-139.7872	90.7232	16.49
1000.2	-10.3279	-143.4589	90.9492	16.53
2000.6	-10.3281	-141.8888	90.9278	16.53
2999.8	-10.3282	-138.4812	89.8963	16.36
5000.1	-10.3288	-140.5748	89.1245	16.23
7001.6	-10.3297	-139.2610	89.5473	16.30
10001.4	-10.3315	-136.4773	87.7685	16.00
20005.9	-10.3425	-132.1135	83.8247	15.35
49984.7	-10.4180	-123.1780	67.9737	12.73
100023.7	-10.6786	-114.6712	63.8714	12.09
150139.1	-11.0878	-109.0131	56.9281	11.01
200342.4	-11.6204	-103.9734	54.9299	10.76

(a) Channel 0 - Input amplitude: -10 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3248	-139.4231	90.5206	16.46
124.9	-10.3248	-139.2064	90.4800	16.45
250.1	-10.3248	-142.3653	90.4757	16.45
312.6	-10.3249	-141.0258	90.4757	16.45
625.1	-10.3249	-143.3590	90.2848	16.42
750.1	-10.3249	-142.5945	90.3559	16.43
1000.2	-10.3253	-143.7360	91.0859	16.55
2000.6	-10.3255	-145.6543	91.0536	16.55
2999.8	-10.3256	-141.5589	90.0001	16.37
5000.1	-10.3262	-140.9901	89.2115	16.24
7001.6	-10.3271	-141.6612	89.6435	16.31
10001.4	-10.3290	-139.0747	87.8330	16.01
20005.9	-10.3400	-132.7200	83.8488	15.35
49984.7	-10.4158	-123.2868	67.9752	12.73
100023.7	-10.6777	-114.9049	63.8718	12.09
150139.1	-11.0878	-109.0131	56.9281	11.01
200342.4	-11.6204	-103.9734	54.9299	10.76

(b) Channel 1 - Input amplitude: -10 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-6.1777	-138.1475	94.7704	16.48
124.9	-6.1778	-140.1304	94.7530	16.47
250.1	-6.1778	-138.8907	94.6614	16.46
312.6	-6.1778	-139.8234	94.6312	16.45
625.1	-6.1779	-140.8151	94.6826	16.46
750.1	-6.1779	-143.3135	94.6984	16.46
1000.2	-6.1782	-140.5183	94.8515	16.49
2000.6	-6.1783	-140.3205	94.7693	16.48
2999.8	-6.1785	-137.3779	93.5392	16.27
5000.1	-6.1790	-138.6529	91.6455	15.96
7001.6	-6.1799	-138.4645	92.2605	16.06
10001.4	-6.1818	-134.8157	85.2125	14.89
20005.9	-6.1927	-129.0484	80.7307	14.15
49984.7	-6.2678	-122.4023	65.4840	11.63
100023.7	-6.5271	-113.7974	60.0172	10.76
150139.3	-6.9355	-105.7599	54.0670	9.84
200342.7	-7.4648	-100.6887	54.9759	10.08

(c) Channel 0 - Input amplitude: -6 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-6.1751	-139.7374	94.5781	16.44
124.9	-6.1752	-141.0688	94.5627	16.44
250.1	-6.1752	-146.0805	94.4963	16.43
312.6	-6.1752	-141.6255	94.3792	16.41
625.1	-6.1753	-144.2615	94.4658	16.43
750.1	-6.1753	-144.4215	94.4962	16.43
1000.2	-6.1756	-144.5874	94.9819	16.51
2000.6	-6.1758	-141.3863	94.9055	16.50
2999.8	-6.1759	-139.2711	93.6425	16.29
5000.1	-6.1765	-138.7918	91.7108	15.97
7001.6	-6.1773	-136.7910	92.3337	16.07
10001.4	-6.1792	-134.9007	85.2307	14.89
20005.9	-6.1902	-128.9451	80.7401	14.15
49984.7	-6.2657	-122.0063	65.4842	11.63
100023.7	-6.5263	-113.5410	60.0174	10.76
150139.3	-6.9355	-105.7599	54.0670	9.84
200342.7	-7.4648	-100.6887	54.9759	10.08

(d) Channel 1 - Input amplitude: -6 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-3.1889	-137.5936	97.5472	16.44
124.9	-3.1889	-138.7869	97.6001	16.45
250.1	-3.1890	-135.9334	97.4721	16.43
312.6	-3.1890	-137.7940	97.4440	16.42
625.1	-3.1891	-139.2680	97.4292	16.42
750.1	-3.1891	-141.6155	97.3211	16.40
1000.2	-3.1893	-142.7647	97.5746	16.45
2000.6	-3.1895	-141.1486	97.3497	16.41
3000.0	-3.1896	-136.3145	92.4491	15.59
5000.1	-3.1902	-137.1042	91.6570	15.46
7001.6	-3.1910	-137.6669	90.2377	15.23
10001.4	-3.1929	-134.6620	91.4707	15.43
20005.9	-3.2038	-128.4541	85.2801	14.41
49984.7	-3.2789	-120.5113	70.0593	11.89
100023.7	-3.5380	-109.6609	62.6702	10.71
150139.3	-3.9469	-102.7287	61.8043	10.63
200342.4	-4.4755	-99.1662	53.1734	9.28

(e) Channel 0 - Input amplitude: -3 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-3.1863	-143.4326	97.3718	16.41
124.9	-3.1864	-141.2826	97.4592	16.43
250.1	-3.1864	-141.3566	97.3509	16.41
312.6	-3.1864	-142.7291	97.3393	16.41
625.1	-3.1865	-142.1296	97.3075	16.40
750.1	-3.1865	-142.7677	97.2071	16.38
1000.2	-3.1867	-142.9264	97.7044	16.47
2000.6	-3.1869	-141.4131	97.4787	16.43
3000.0	-3.1870	-138.0288	92.4940	15.60
5000.1	-3.1876	-135.7210	91.6943	15.47
7001.6	-3.1885	-136.5631	90.2596	15.23
10001.4	-3.1903	-133.3305	91.5013	15.44
20005.9	-3.2013	-128.3579	85.2855	14.41
49984.7	-3.2768	-120.3264	70.0592	11.89
100023.7	-3.5372	-109.3809	62.6702	10.71
150139.3	-3.9469	-102.7287	61.8043	10.63
200342.4	-4.4755	-99.1662	53.1734	9.28

(f) Channel 1 - Input amplitude: -3 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-1.1749	-134.9556	99.4394	16.42
124.9	-1.1749	-137.6100	99.4282	16.42
250.1	-1.1750	-134.3593	99.1165	16.37
312.6	-1.1750	-134.6656	99.2057	16.38
625.1	-1.1751	-137.5868	99.1823	16.38
750.1	-1.1751	-138.0458	99.1500	16.37
1000.2	-1.1755	-137.9632	99.1557	16.37
2000.6	-1.1756	-139.3988	98.5262	16.27
2999.8	-1.1758	-136.1986	93.0746	15.36
5000.1	-1.1763	-135.0105	88.3519	14.58
7001.6	-1.1772	-135.1210	91.6023	15.12
10001.4	-1.1790	-133.6586	86.1879	14.22
20005.9	-1.1899	-128.8402	85.0081	14.03
49984.7	-1.2650	-119.0961	66.0501	10.89
100024.0	-1.5241	-108.3697	62.3577	10.32
150139.3	-1.9324	-101.9871	59.7903	9.96
200342.4	-2.4606	-99.0985	60.5827	10.18

(g) Channel 0 - Input amplitude: -1 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-1.1723	-142.4291	99.2352	16.39
124.9	-1.1724	-142.2236	99.2736	16.39
250.1	-1.1724	-138.4414	98.7700	16.31
312.6	-1.1724	-140.0544	99.0136	16.35
625.1	-1.1725	-141.9501	99.0730	16.36
750.1	-1.1725	-141.8365	99.0082	16.35
1000.2	-1.1729	-143.7667	99.2705	16.39
2000.6	-1.1730	-143.8074	98.6325	16.29
2999.8	-1.1732	-136.8253	93.1098	15.37
5000.1	-1.1737	-135.3874	88.3603	14.58
7001.6	-1.1746	-135.3357	91.6294	15.12
10001.4	-1.1764	-133.8301	86.1944	14.22
20005.9	-1.1874	-128.8034	85.0111	14.03
49984.7	-1.2629	-118.9272	66.0502	10.89
100024.0	-1.5234	-108.2275	62.3575	10.32
150139.3	-1.9324	-101.9871	59.7903	9.96
200342.4	-2.4606	-99.0985	60.5827	10.18

(h) Channel 1 - Input amplitude: -1 dBFS

2.3 Channels averaging

In this subsection the results when the two channels of each single EVAL board are averaged in the time domain and in the frequency domain. The data was capture in eight frames of 2^{23} each, resulting in a total number of samples equal to 67108864 per channel. Then, the complete dataset was divided in 8 frames of 2^{23} samples and processed.

Table 2: Channel Averaging on Boards - Results

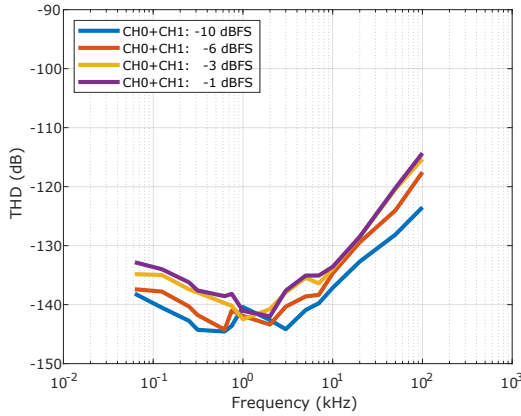
Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3275	-138.1483	98.2416	17.74
124.9	-10.3275	-140.5098	98.2101	17.74
250.1	-10.3277	-142.7449	98.1281	17.72
312.6	-10.3277	-144.2712	98.0969	17.72
625.1	-10.3281	-144.5632	98.0332	17.71
750.1	-10.3284	-143.6037	98.0972	17.72
1000.2	-10.3290	-140.4149	97.9480	17.69
2000.6	-10.3329	-142.6158	97.8744	17.68
2999.8	-10.3392	-144.1702	92.9368	16.86
5000.1	-10.3595	-140.9139	91.5445	16.64
7001.6	-10.3899	-139.7767	88.3032	16.10
10001.4	-10.4536	-137.1576	90.0489	16.40
20005.9	-10.8121	-132.7095	82.4366	15.20
49984.7	-12.7577	-128.1310	62.2143	12.16
100023.7	-16.5566	-123.4995	61.1841	12.62

(a) Board RA Channel averaging - Input amplitude: -10 dBFS

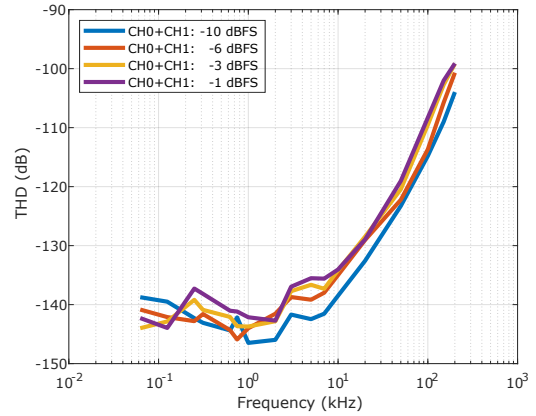
Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3261	-138.7425	93.9870	17.04
124.9	-10.3261	-139.5011	93.9783	17.03
250.1	-10.3261	-142.2927	93.8712	17.02
312.6	-10.3261	-143.0861	93.9045	17.02
625.1	-10.3262	-144.3692	93.9213	17.02
750.1	-10.3262	-142.1688	93.9011	17.02
1000.2	-10.3266	-146.4886	93.8043	17.01
2000.6	-10.3268	-145.9718	93.7531	17.00
2999.8	-10.3269	-141.6913	91.9559	16.70
5000.1	-10.3275	-142.4633	90.7778	16.50
7001.6	-10.3284	-141.5461	91.4083	16.61
10001.4	-10.3302	-138.4836	88.9177	16.19
20005.9	-10.3413	-132.5670	84.2511	15.42
49984.7	-10.4169	-123.2798	67.9849	12.73
100023.7	-10.6782	-114.7945	63.8759	12.09
150139.1	-11.0878	-109.0131	56.9281	11.01
200342.4	-11.6204	-103.9734	54.9299	10.76

(b) Board RI Channel averaging - Input amplitude: -10 dBFS

Figure 3: Boards Channel averaging - THD vs Input signal amplitude and frequency



(a) Board RA - THD - Channels averaging - Dataset no. 2



(b) Board RI - THD - Channels averaging - Dataset no. 2

2.4 Channels and FFT averaging

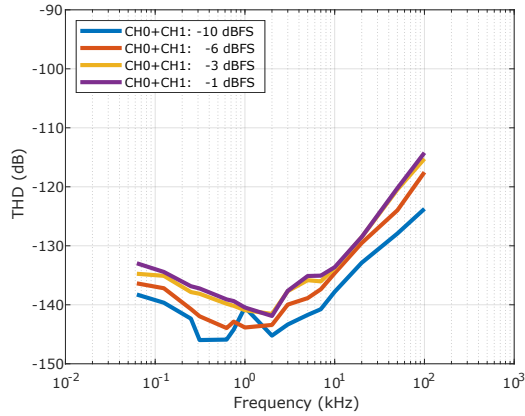
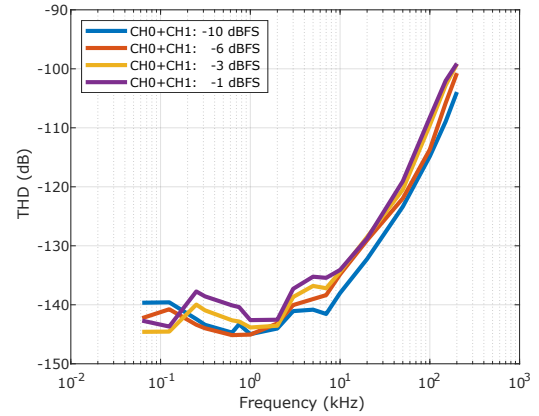
Total harmonic distortion when averaging the ADC two channels.

Table 3: Channel and FFT averaging on Boards - Results

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3275	-138.2632	99.1539	17.89
124.9	-10.3275	-139.6522	99.1128	17.89
250.1	-10.3277	-142.3525	98.9926	17.87
312.6	-10.3277	-145.9773	98.9723	17.86
625.1	-10.3282	-145.9010	98.9365	17.86
750.1	-10.3284	-144.1997	98.9760	17.86
1000.2	-10.3290	-140.4557	98.8244	17.84
2000.6	-10.3329	-145.2261	98.6656	17.81
3000.0	-10.3392	-143.3427	94.1246	17.06
5000.1	-10.3595	-141.7181	91.9007	16.69
7001.6	-10.3899	-140.7539	91.5995	16.65
10001.4	-10.4536	-137.7956	89.9175	16.38
20005.9	-10.8121	-132.8764	85.8722	15.77
49984.7	-12.7588	-127.9222	66.4410	12.86
100023.7	-16.5567	-123.7358	64.1636	13.12

(a) Board RA CHs and FFT averaging - Input amplitude: -10 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3261	-139.6545	94.8948	17.19
124.9	-10.3261	-139.5839	94.8881	17.19
250.1	-10.3261	-142.4242	94.8383	17.18
312.6	-10.3262	-143.3949	94.8201	17.17
625.1	-10.3262	-144.7155	94.7950	17.17
750.1	-10.3262	-143.3017	94.8185	17.17
1000.2	-10.3266	-144.9764	94.7257	17.16
2000.6	-10.3268	-144.0188	94.6504	17.15
3000.0	-10.3269	-141.0726	92.3156	16.76
5000.1	-10.3275	-140.8297	90.7702	16.50
7001.6	-10.3284	-141.5555	90.3169	16.43
10001.4	-10.3302	-138.0081	89.1645	16.24
20005.9	-10.3413	-132.2358	85.3480	15.60
49984.7	-10.4182	-123.3692	66.4914	12.48
100023.7	-10.6782	-114.8296	64.0761	12.13
150139.1	-11.0894	-108.9273	56.9282	11.01
200342.4	-11.6229	-103.9398	54.9302	10.76

(b) Board RI CHs and FFT averaging - Input amplitude: -10 dBFS**Figure 4:** Boards Channel and FFT averaging - THD vs Input signal amplitude and frequency**(a)** Board RA-THD-Channels and FFT averaging**(b)** Board RI-THD-Channels and FFT averaging

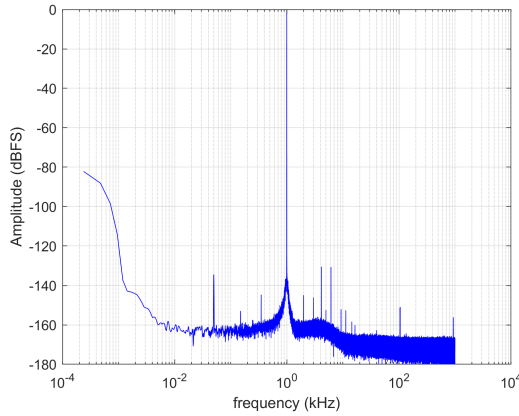
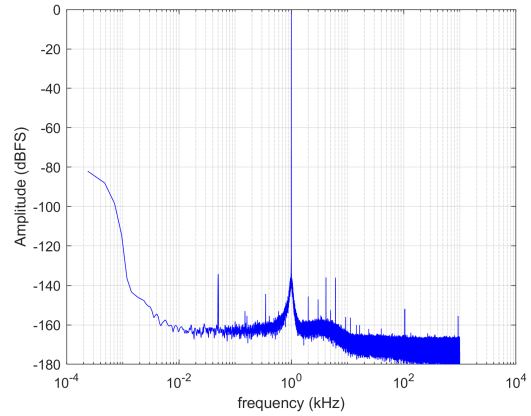
Signal spectrum when averaging the FFT results over all windows datasets (8 datasets in total)

Table 4: Channel and FFT averaging on Boards - Results

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3275	-138.2632	99.1539	17.89
124.9	-10.3275	-139.6522	99.1128	17.89
250.1	-10.3277	-142.3525	98.9926	17.87
312.6	-10.3277	-145.9773	98.9723	17.86
625.1	-10.3282	-145.9010	98.9365	17.86
750.1	-10.3284	-144.1997	98.9760	17.86
1000.2	-10.3290	-140.4557	98.8244	17.84
2000.6	-10.3329	-145.2261	98.6656	17.81
3000.0	-10.3392	-143.3427	94.1246	17.06
5000.1	-10.3595	-141.7181	91.9007	16.69
7001.6	-10.3899	-140.7539	91.5995	16.65
10001.4	-10.4536	-137.7956	89.9175	16.38
20005.9	-10.8121	-132.8764	85.8722	15.77
49984.7	-12.7588	-127.9222	66.4410	12.86
100023.7	-16.5567	-123.7358	64.1636	13.12

(a) Board RA CHs and FFT averaging - Input amplitude: -10 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3261	-139.6545	94.8948	17.19
124.9	-10.3261	-139.5839	94.8881	17.19
250.1	-10.3261	-142.4242	94.8383	17.18
312.6	-10.3262	-143.3949	94.8201	17.17
625.1	-10.3262	-144.7155	94.7950	17.17
750.1	-10.3262	-143.3017	94.8185	17.17
1000.2	-10.3266	-144.9764	94.7257	17.16
2000.6	-10.3268	-144.0188	94.6504	17.15
3000.0	-10.3269	-141.0726	92.3156	16.76
5000.1	-10.3275	-140.8297	90.7702	16.50
7001.6	-10.3284	-141.5555	90.3169	16.43
10001.4	-10.3302	-138.0081	89.1645	16.24
20005.9	-10.3413	-132.2358	85.3480	15.60
49984.7	-10.4182	-123.3692	66.4914	12.48
100023.7	-10.6782	-114.8296	64.0761	12.13
150139.1	-11.0894	-108.9273	56.9282	11.01
200342.4	-11.6229	-103.9398	54.9302	10.76

(b) Board RI CHs and FFT averaging - Input amplitude: -10 dBFS**Figure 5:** FFT averaging - Output signal spectrum for a -1 dBFS, 1 kHz input signal**(a)** Board RA-FFT averaging output spectrum**(b)** Board RI-FFT averaging output spectrum

3 Parallel Operation - Two boards, four channels

Two AD4630-24 evaluation boards in parallel operation 4 channels post-processing time domain averaging.

3.1 Dynamic parameters results

In this subsection the formula and simple definition are explained.

THD: Total harmonic distortion. Ratio between the r.m.s. amplitude of harmonic contents to the fundamental r.m.s amplitude, according to:

$$THD \text{ (dB)} = 20 \cdot \log_{10} \left(\frac{\sqrt{\sum_{j=1}^N A_j^2}}{A_0} \right) \quad (2)$$

SINAD: Signal-to.noise ratio plus distortion

$$SINAD \text{ (dB)} = \frac{S}{N} + THD \quad (3)$$

ENOB: Effective Number of Bits, considering input signal amplitude lower than FS

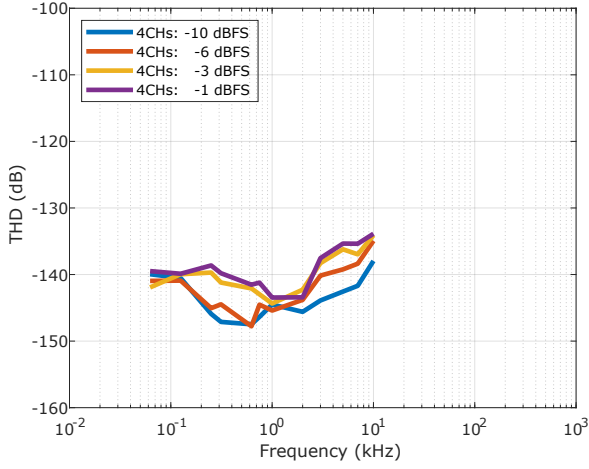
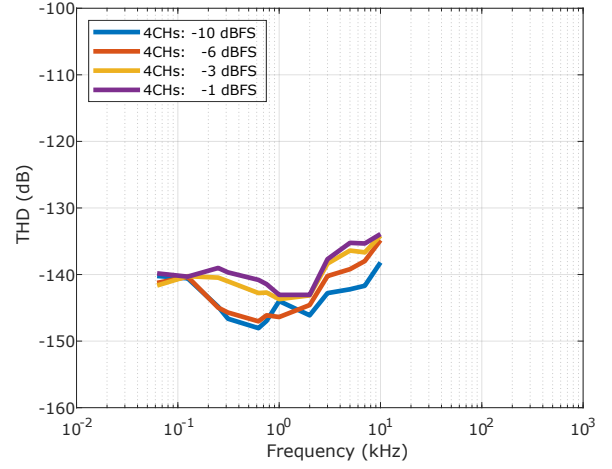
$$ENOB = \frac{SINAD - 1.76 \text{ dB} + \text{Input level below FS}}{6.02 \text{ dB}} \quad (4)$$

Table 5: 2 Boards 4 Channel in Parallel Dataset and FFT averaging - Results

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3268	-139.9553	98.6113	17.80
124.9	-10.3268	-140.5213	98.6159	17.80
250.1	-10.3269	-145.9094	98.5593	17.80
312.6	-10.3269	-147.1177	98.4753	17.78
625.1	-10.3272	-147.4747	98.4855	17.78
750.1	-10.3273	-146.3566	98.5421	17.79
1000.2	-10.3278	-144.5083	98.4263	17.77
2000.6	-10.3298	-145.6047	98.3400	17.76
2999.8	-10.3331	-143.8960	95.4755	17.28
5000.1	-10.3435	-142.5620	94.7991	17.17
7001.6	-10.3591	-141.6890	93.1492	16.90
10001.4	-10.3917	-137.9586	91.6424	16.66

(a) 2 Boards 4-Chs Dataset averaging -
Dataset no. 2 - Input amplitude: -10 dBFS

Freq. (Hz)	Fund. (dBFS)	THD (dB)	SINAD (dB)	ENOB
62.5	-10.3268	-140.2196	99.5188	17.95
124.9	-10.3268	-140.5708	99.5029	17.95
250.1	-10.3269	-144.8356	99.4375	17.94
312.6	-10.3269	-146.6421	99.4131	17.94
625.1	-10.3272	-148.0534	99.3732	17.93
750.1	-10.3273	-146.9779	99.4202	17.94
1000.2	-10.3278	-143.9858	99.3157	17.92
2000.6	-10.3298	-146.1256	99.1886	17.90
3000.0	-10.3344	-142.7917	95.9996	17.37
5000.1	-10.3435	-142.2183	94.2281	17.08
7001.6	-10.3591	-141.6884	94.1233	17.06
10001.4	-10.3918	-138.1884	92.0571	16.73

(b) 2 Boards 4-Chs FFT averaging -
Input amplitude: -10 dBFS**Figure 6:** THD for parallel boards operation and overall channels averaging in frequency range from 62.5 Hz to 10 kHz.**(a)** Parallel channels operation and averaging - Data set time domain averaging**(b)** Parallel channels operation and averaging - Time domain and FFT averaging

3.2 ENOB calculation and comparison

In order to improve the SNR (and potentially also other parameters) by parallel ADC operation, multiple ADCs are followed by post processing to digital summation of their outputs. Using this simple technique, the perfectly correlated signals in the output sum result will increase by a factor N , while perfectly uncorrelated signals will only increase by a factor of $N/2$.

In the case of adding the input voltage signal from four different ADC, the output common signal, V_{TOT} after post processing results in the linear summation of the input voltage signal, V_{ADC_j} , from each ADC, V_{in} , plus the root square sum of the noise voltages, e_{N_j} , as [3],

$$V_{TOT} = \sum_{j=1}^M V_{ADC_j} + \sqrt{\sum_{j=1}^m e_{N_j}^2}, \quad (5)$$

where M is the total number of ADC (4 in this case).

Since $\text{SNR}(\text{dB}) = 6.02 N + 1.76$, where N is the number of bits, the above assumption gives the increase in effective resolution:

$$N + \Delta N = \frac{1}{6.02} \cdot (\text{SNR}(\text{dB}) - 1.76 + 6.02 \text{ dB}) = \left(\frac{\text{SNR}(\text{dB})}{6.02} - \frac{1.76}{6.02} \right) + \frac{6.02 \text{ dB}}{6.02} = N + 1 \quad (6)$$

The Effective Number of Bits was calculated for measurements setup and conditions. Then, it is compared in this section. The comparison is made at reference frequency of 1 kHz. Table shows the results and comparison.

Table 6: Parallel Operation - Results - Last columns show the ΔN when averaging 4-CH from 1-CH

Board	Meas Type	Ch	Freq (Hz)	Fund (dBFS)	SINAD (dB)	ENOB	ΔN
RI+RA	2B-4CH-FFT AVG	4	1000.2	-10.32	99.31	17.92	1.37
RI+RA	2B-4CH-DS AVG	4			98.42	17.77	
RI	1B CH+FFT_AVG	2			94.72	17.16	
RI	1B CH_AVG DS	2			93.80	17.01	
RI	Single channel	0			90.94	16.53	
RI	Single channel	1			91.08	16.55	

(a) Results for input signal amplitude -10 dBFS and frequency 1 kHz. The $\Delta N = 1.37$ bits

Ch	Freq (Hz)	Fund (dBFS)	SINAD (dB)	ENOB	ΔN
4	1000.2	-6.17	102.89	17.83	1.31
4			102.06	17.69	
2			98.39	17.08	
2			97.50	16.93	
0			94.85	16.49	
1			94.98	16.51	

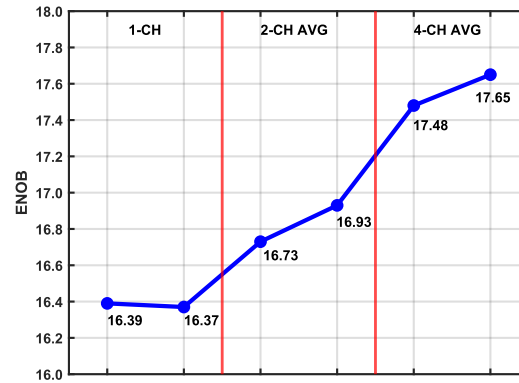
(b) Results for input signal amplitude -6 dBFS and frequency 1 kHz. The $\Delta N = 1.31$ bits

Ch	Freq (Hz)	Fund (dBFS)	SINAD (dB)	ENOB	ΔN
4	1000.2	-3.18	105.23	17.72	1.25
4			104.41	17.58	
2			100.84	16.99	
2			100.00	16.85	
0			97.57	16.45	
1			97.70	16.47	

(c) Results for input signal amplitude -3 dBFS and frequency 1 kHz. The $\Delta N = 1.25$ bits

Ch	Freq (Hz)	Fund (dBFS)	SINAD (dB)	ENOB	ΔN
4	1000.2	-1.17	106.84	17.65	1.26
4			105.84	17.48	
2			102.49	16.93	
2			101.29	16.73	
0			99.16	16.37	
1			99.27	16.39	

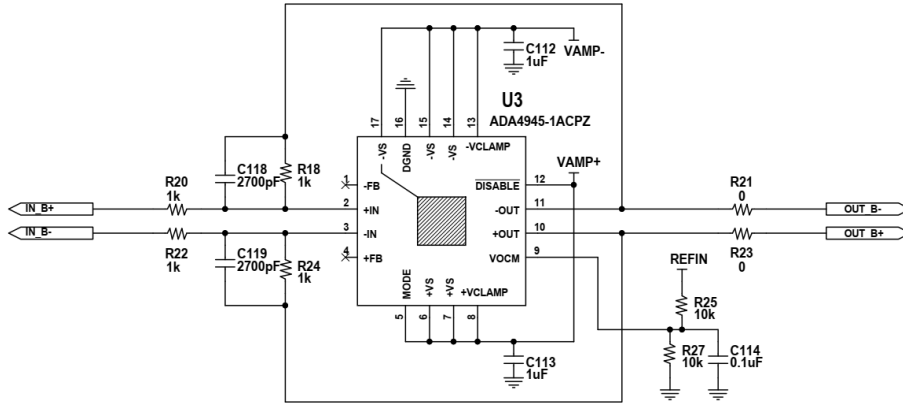
(d) Results for input signal amplitude -1 dBFS and frequency 1 kHz. The $\Delta N = 1.26$ bits

Figure 7: Parallel channels operation and averaging - Effective Number of Bits for input amplitude -1.17 dBFS.

4 Board RI bandwidth extension

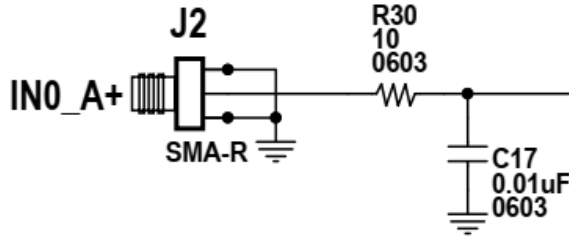
The EVAL Board comes with different possible front-end configurations, see ADI web page, the default is a fully differential amplifier with $G=1$ and a low pass filter having a cut-off frequency around $f_{-3\text{ dB}} = 54\text{ kHz}$.

Figure 8: Fully differential amplifier circuit at the EVAL Board.
R18-C118 and R24-C119 conform a LPF with $f_{-3\text{ dB}} = 54\text{ kHz}$



Input LPF is

Figure 9: Input LPF with a $f_{-3\text{ dB}} = 1.59\text{ MHz}$

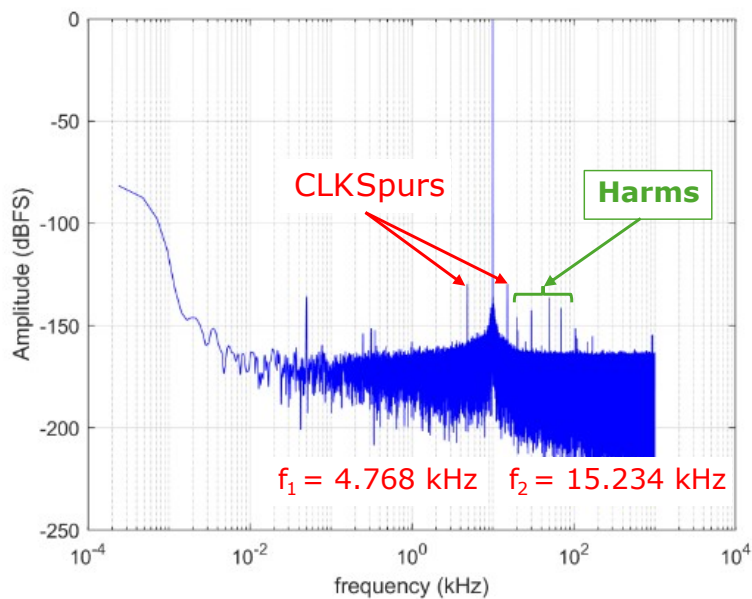


5 Sampling clock spurs frequency

Clock spurs appeared in the spectrum [4], which are mainly due to jitter at the FPGA of the evaluation board. As depicted in figure 10 there are two images at:

$$f = \begin{cases} f_1 = f_{IN} + (f_{spur} - f_{CLK}) \\ f_2 = f_{IN} - (f_{spur} - f_{CLK}) \end{cases}$$

where f_{IN} is the input signal frequency, f_{spur} is the spur frequency and f_{CLK} is the sampling clock. For an input signal frequency of 10 kHz, a sampling clock of 2 MHz, the spurs are at $f = 1.994766\text{ MHz}$, which gives a Δf equal to 5232 Hz.

Figure 10: Clock Spurs

References

- [1] AD4630-24 SAR ADC Evaluation Board, Analog Devices inc.
- [2] F. J. Harris, "On the use of windows for harmonic analysis with the discrete Fourier transform," in Proceedings of the IEEE, vol. 66, no. 1, pp. 51-83, Jan. 1978.
- [3] Rob Reeder, Mark Looney and Jim Hand, "Pushing the State of the Art with Multichannel A/D Converters", Analog Dialogue, Vol.39, No. 05, May 2005.
- [4] Thomas Neu, "Impact of sampling-clock spurs on ADC performance ", Analog Applications Journal, Texas Instruments Incorporated, 3Q 2009.